Tutorial

Modern Physics

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For exercises from the textbook, the corresponding number is reported. Text may have been edited, for clarity. Unless otherwise specified, all results are given with three significant digits, avoiding rounding errors.

The solutions are provided on an "as is" basis, without warranty. Any error report is appreciated.

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WEEK 14

PARTICLE PHYSICS

- **Plane wave solutions of the Dirac equation**
  In the lecture we wrote down the Dirac equation

$$(i\hbar\not{\partial} - mc^2)\,\psi(x) = 0,$$

where $\not{\partial} = \sum_{\mu=0}^{3} \gamma_\mu \partial_\mu$ is a matrix valued differential operator. We use the indexing convention that the zeroth spacetime component corresponds to time $\partial_0 = \partial_t$, and $\mu = 1, 2, 3$ corresponds to x, y, z respectively with a factor of c, $\partial_1 = c\partial_x$, etc. The γ matrices are conventionally defined as (written in a $2+2$ block form)

$$\gamma_0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \qquad \gamma_i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix} \text{ for } i = 1, 2, 3,$$

where σ_i are the Pauli matrices and I_2 is the 2×2 identity matrix. The key property of these matrices that they satisfy the Dirac algebra:

$$\{\gamma_\mu, \gamma_\nu\} = \gamma_\mu\gamma_\nu + \gamma_\nu\gamma_\mu = \begin{cases} 2I_4 & \text{if } \mu = \nu = 0, \\ -2I_4 & \text{if } \mu = \nu \neq 0, \\ 0 & \text{otherwise.} \end{cases}$$

The wavefunction ψ is a four-component complex function (a four-spinor), that depends on the spacetime coordinate $x = (t, \mathbf{x})$.

(a) Show that it is possible to write the Dirac equation in an equivalent form similar to Schrödinger's equation:

$$i\hbar\partial_t\psi(x) = \left(-i\hbar\sum_{i=1}^{3}\alpha_i\partial_i + mc^2\alpha_0\right)\psi(x), \tag{1}$$

where $\alpha_i = \gamma_0\gamma_i$ for $i = 1, 2, 3$ and $\alpha_0 = \gamma_0$. Show that the α matrices square to 1 and mutually anticommute, i.e. $\{\alpha_\mu, \alpha_\nu\} = 2\delta_{\mu\nu} I_4$ for $\mu, \nu = 0, 1, 2, 3$.

(b) Write equation (1) explicitly in a 4×4 matrix-vector form using the differential operators $\partial_t, \partial_x, \partial_y, \partial_z$ and the components of the wavefunction $\psi = (\psi_1, \psi_2, \psi_3, \psi_4)$.

(c) We look for a solution in the form of a plane-wave:

$$\psi(x) = \psi(t, \mathbf{x}) = u_{\mathbf{k}} e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})},$$

where ω and $\mathbf{k}$ are the angular frequency and wavenumber of the propagating wave, and u is a four-component vector that only depends on $\mathbf{k}$, but not on t and $\mathbf{x}$, describing its polarization (e.g. spin). Substitute the plane-wave ansatz into equation (1). Using the properties of the exponential function you should find that the result has the form $\hbar\omega\, u_{\mathbf{k}} e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})} = H_{\mathbf{k}} u_{\mathbf{k}} e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})}$ where $H_{\mathbf{k}}$ is a 4×4 matrix. (Instead of working explicitly in terms of 4×4 matrices, it is easier to write the matrix in terms of the α matrices.)

(d) In order to solve the Dirac equation, $\hbar\omega\, u_{\mathbf{k}} = H_{\mathbf{k}} u_{\mathbf{k}}$ needs to be satisfied, which is an eigenvalue-eigenvector equation for $\hbar\omega$ and u. Show that this equation imposes constraint between ω and $\mathbf{k}$ that is equivalent to the relativistic energy-momentum relation using $E = |\hbar\omega|$ and $\mathbf{p} = \hbar\mathbf{k}$. (Use that the eigenvalues of the square of a matrix are the squares of the eigenvalues, and use the commutation properties of the α matrices to simplify the square.)

(e) Find the four eigenvalues ω_i and eigenvectors u_i at $\mathbf{k} = 0$, and give the physical interpretation of each.

Solution

(a) It is straightforward to check that

$$\gamma_0^2 = I_4, \qquad \alpha_i = \gamma_0\gamma_i = \begin{pmatrix} 0 & \sigma_i \\ \sigma_i & 0 \end{pmatrix} \text{ for } i = 1, 2, 3,$$

as well as the anticommutation relations. Multiplying the equation $(i\hbar\partial\!\!\!/ - mc^2)\ \psi(x) = 0$ by γ_0 from the left we get

$$0 = \gamma_0(i\hbar\partial\!\!\!/ - mc^2)\ \psi(x) = (i\hbar\gamma_0^2\partial_t + i\hbar\sum_{i=1}^{3}\gamma_0\gamma_i\partial_i - \gamma_0 mc^2)\ \psi(x) = (i\hbar\partial_t + i\hbar\sum_{i=1}^{3}\alpha_i\partial_i - \alpha_0 mc^2)\ \psi(x).$$

Rearranging the equation gives (1). As γ_0 is unitary, the two equations are equivalent, the same wavefunction solve both.

Note: we can rewrite the matrices appearing here in a tensor product form: $\gamma_0 = \alpha_0 = \sigma_z \otimes I_2$, $\gamma_i = -i\sigma_y \otimes \sigma_i$, and $\alpha_i = \sigma_x \otimes \sigma_i$. For tensor product operators (anti)commutation relations can be evaluated term-by term, allowing simple calculations. In quantum computation science logical operations performed by the computer are often represented as tensor products of Pauli matrices.

(b) Writing equation (1) explicitly:

$$\begin{pmatrix} i\hbar\partial_t\psi_1 \\ i\hbar\partial_t\psi_2 \\ i\hbar\partial_t\psi_3 \\ i\hbar\partial_t\psi_4 \end{pmatrix} = \begin{pmatrix} mc^2 & 0 & -i\hbar c\partial_z & -i\hbar c(\partial_x - i\partial_y) \\ 0 & mc^2 & -i\hbar c(\partial_x + i\partial_y) & i\hbar c\partial_z \\ -i\hbar c\partial_z & -i\hbar c(\partial_x - i\partial_y) & -mc^2 & 0 \\ -i\hbar c(\partial_x + i\partial_y) & i\hbar c\partial_z & 0 & -mc^2 \end{pmatrix} \begin{pmatrix} \psi_1 \\ \psi_2 \\ \psi_3 \\ \psi_4 \end{pmatrix},$$

$$\begin{pmatrix} i\hbar\partial_t\psi_1 \\ i\hbar\partial_t\psi_2 \\ i\hbar\partial_t\psi_3 \\ i\hbar\partial_t\psi_4 \end{pmatrix} = \begin{pmatrix} mc^2\psi_1 - i\hbar c(\partial_z\psi_3 - \partial_x\psi_4 - i\partial_y\psi_4) \\ mc^2\psi_2 - i\hbar c(\partial_x\psi_3 + i\partial_y\psi_3 - \partial_z\psi_4) \\ -mc^2\psi_3 - i\hbar c(\partial_z\psi_1 + \partial_x\psi_2 - i\partial_y\psi_2) \\ -mc^2\psi_4 - i\hbar c(\partial_x\psi_1 + i\partial_y\psi_1 - \partial_z\psi_2) \end{pmatrix},$$

which is a coupled system of partial differential equations.

(c) Using that $i\partial_t e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})} = \omega e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})}$ and $-i\partial_i e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})} = ck_i e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})}$ we substitute the plane-wave ansatz into (1):

$$\hbar\omega\ u_{\mathbf{k}} e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})} = \left(\hbar c\sum_{i=1}^{3}\alpha_i k_i + mc^2\alpha_0\right) u_{\mathbf{k}} e^{-i(\omega t - \mathbf{k}\cdot\mathbf{x})}.$$

This way we can identify $H_{\mathbf{k}} = \hbar c\sum_{i=1}^{3}\alpha_i k_i + mc^2\alpha_0$.

(d) As the equation needs to be true for all t and $\mathbf{x}$, we can simplify by the exponential to obtain the eigenvalue equation

$$\hbar\omega\ u_{\mathbf{k}} = \left(\hbar c\sum_{i=1}^{3}\alpha_i k_i + mc^2\alpha_0\right) u_{\mathbf{k}} = H_{\mathbf{k}} u_{\mathbf{k}}.$$

Taking the square of $H_{\mathbf{k}}$:

$$H_{\mathbf{k}}^2 = \left(\hbar c\sum_{i=1}^{3}\alpha_i k_i + mc^2\alpha_0\right)^2 = \sum_{i=1}^{3}(\hbar c k_i)^2 + m^2c^4 = c^2\hbar^2\mathbf{k}^2 + m^2c^4,$$

where we used the anticommutation relation $\alpha_\mu\alpha_\nu = -\alpha_\nu\alpha_\mu$ for $\mu \neq \nu$, meaning that all the cross-terms in the square cancel, and that $\alpha_\mu^2 = I_4$ (it is customary to omit the identity matrix I_4). As $H_{\mathbf{k}}^2$ is proportional to the identity matrix, it has the fourfold degenerate eigenvalue $c^2\hbar^2\mathbf{k}^2 + m^2c^4$. This means that the eigenvalues of $H_{\mathbf{k}}$, $\hbar\omega$ satisfy

$$(\hbar\omega)^2 = c^2\hbar^2\mathbf{k}^2 + m^2c^4 \quad \Rightarrow \quad E^2 = c^2\mathbf{p}^2 + m^2c^4.$$

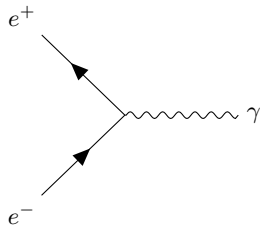
(e) Substituting $\mathbf{k} = 0$, $H_{\mathbf{k}}$ becomes diagonal:

$$H_{\mathbf{k}=0} = mc^2\alpha_0 = \begin{pmatrix} mc^2 & 0 & 0 & 0 \\ 0 & mc^2 & 0 & 0 \\ 0 & 0 & -mc^2 & 0 \\ 0 & 0 & 0 & -mc^2 \end{pmatrix},$$

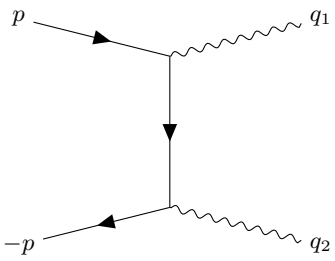
with a pair of twofold degenerate eigenvalues of $+mc^2$ and $-mc^2$, and the eigenvectors are the four unit vectors. It is true in general that $H_{\mathbf{k}}$ has a pair of twofold degenerate eigenvalues of $\hbar\omega = \pm\sqrt{\hbar^2c^2\mathbf{k}^2 + m^2c^4}$. The two positive eigenvalues correspond to spin up and spin down electrons (particles), and the second two to spin up and spin down positrons (antiparticles). The Dirac equation describes not only electrons, but all massive fermionic particles. The precise interpretation of the negative energy eigenvalue of antiparticles is beyond the scope of this course, but it is important to note that antiparticles have *positive* energy and rest mass, just like particles. The negative result for $\hbar\omega$ rather means that antiparticles oscillate "backwards", explaining the notation in Feynman diagrams that antiparticles propagate "backwards" in time.

- **Kinematics of scattering processes**
  Every scattering process described by an allowed Feynman diagram contributes to the scattering amplitudes. However, the external legs of the diagram need to satisfy kinematic constraints for the process to be allowed. These constraints come from conservation laws, here, we consider conservation of energy and momentum.

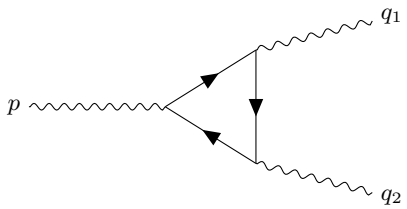
  (a) Consider the diagram describing an electron and positron annihilating and emitting a single photon. Show that this process cannot happen while satisfying energy and momentum conservation.



  (b) Consider the electron-positron annihilation process emitting two photons. Assuming that we are in the center-of-mass frame where the incoming particles have momenta p and $-p$, calculate the momenta q_1 and q_2 of the outgoing photons.



  (c) Consider the diagram describing an incoming photon decaying into two photons. Find the kinematic constraints between the momenta of the incoming photon p and the outgoing photons q_1 and q_2.
  Bonus: This has a solution, however, this process is still forbidden. Can you find another conservation law that cannot be satisfied?



Solution

(a) Electrons and positrons are massive particles and move at speeds lower than c. This means that there is a center-of-mass reference frame where the particles have momenta p and $-p$, such that the total momentum is zero. In this reference frame, the total energy of the particles is

$$E = 2\sqrt{(pc)^2 + (mc^2)^2} > 0.$$

Photons always move at the speed of light, and have the energy-momentum relation

$$E = |p|c.$$

It is clear that a photon with nonzero energy cannot have zero momentum, hence, the conservation laws cannot be satisfied.

(b) The total momentum of the incoming particles is zero, meaning that the outgoing particles also have zero total momentum, $q_2 = -q_1$. The total energy of the particles is

$$E = 2\sqrt{(pc)^2 + (mc^2)^2}.$$

This is converted into the energy of the photons,

$$E = 2|q_1|c.$$

This gives the momentum of the photons

$$|q_1| = |q_2| = \sqrt{p^2 + (mc)^2}.$$

The direction of the outgoing photons is not constrained by these conservation laws.

(c) Momentum conservation implies

$$p = q_1 + q_2,$$

and energy conservation implies

$$E = c|p| = c|q_1| + c|q_2|,$$

which simplifies to

$$|q_1 + q_2| = |q_1| + |q_2|.$$

Recall the *triangle inequality* $|q_1 + q_2| \leq |q_1| + |q_2|$, where the equality is satisfied exactly when the vectors are parallel. So the conservation laws constrain q_1 and q_2 to be parallel to p, and $q_1 + q_2 = p$.
There are several reasons why this process still does not occur. The easiest to explain is conservation of angular momentum. Photons are spin 1 particles, however, because they travel at the speed of light, their angular momentum component in the direction of the linear momentum can only take the values $\pm\hbar$ (and not 0, which is possible for massive spin 1 particles). This is also called *helicity*, and the values $\pm\hbar$ correspond to left and right circularly polarized light. (Linearly polarized light does not correspond to angular momentum 0, it is rather an equal weight superposition of $+\hbar$ and $-\hbar$.) With this in mind, and using the fact that all three photons have the same momentum direction, we find that in order to conserve angular momentum we need to solve $\pm\hbar = \pm\hbar \pm \hbar$, which is impossible as the right hand side can only take the values of 0 and $\pm 2\hbar$.

- **Short-range forces: pion as mediator / Ex. 14.2 (Thorton)**
  The mass of the charged pion is $m_{\pi^+} = 140\,\mathrm{MeV/c^2}$. Determine the range of the nuclear force if the pion is the mediator.

  Solution
  The range of the interaction can be estimated as the *reduced* Compton wavelength of the mediator. In the case of π^+, this is given by

  $$R = \lambda\!\!\!^{-} = \frac{\lambda}{2\pi} = \frac{\hbar}{m_{\pi^+}c} = \frac{\hbar c}{m_{\pi^+}c^2} = 1.41\,\mathrm{fm},$$

  which was obtained using $\hbar c = 197.3 \times 10^{-9}\,\mathrm{eV\,m}$.

- **Short-range forces: Z^0 as a mediator / Ex. 15.5 (Serway)**
  One of the mediators of the weak interaction is the Z^0 boson, which has a mass of $m_{Z^0} = 91\,\mathrm{GeV/c^2}$. Use this information to find an approximate value for the range of the weak interaction.

  Solution
  The range of the interaction can be estimated as the *reduced* Compton wavelength of the mediator. In the case of Z^0, this is given by

  $$R = \lambda\!\!\!^{-} = \frac{\lambda}{2\pi} = \frac{\hbar}{m_{Z^0}c} = \frac{\hbar c}{m_{Z^0}c^2} = 2.17 \times 10^{-18}\,\mathrm{m} = 0.002\,17\,\mathrm{fm},$$

  which was obtained using $\hbar c = 197.3 \times 10^{-9}\,\mathrm{eV\,m}$.

- **Neutrinos from supernovae / Exs. 15.23 (Serway)**
  The most recent naked-eye supernova was Supernova Shelton 1987A (Fig. P15.23). It was 170 000 ly away in the next galaxy to ours, the Large Magellanic Cloud. About 3 h before its optical brightening was noticed, two continuously running neutrino detection experiments simultaneously registered the first neutrinos from an identified source other than the Sun. The *Irvine-Michigan-Brookhaven* experiment in a salt mine in Ohio registered $n_\nu^{\mathrm{IMB}} = 8$ neutrinos over a $\Delta t = 6\,\mathrm{s}$ period, and the *Kamiokande II* experiment in a zinc mine in Japan counted $n_\nu^{\mathrm{KII}} = 11$ neutrinos in $\Delta t = 13\,\mathrm{s}$. (Because the supernova is far south in the sky, these neutrinos entered the detectors from below. They passed through the Earth before they were by chance absorbed by nuclei in the detectors.) The neutrino energies were between about $E_\nu^{\mathrm{IMB}} = 8\,\mathrm{MeV}$ and $E_\nu^{\mathrm{KII}} = 40\,\mathrm{MeV}$. If neutrinos have no mass, then neutrinos of all energies should travel together at the speed of light —the data are consistent with this possibility. The arrival times could show scatter simply because neutrinos were created at different moments as the core of the star collapsed into a neutron star. If neutrinos have nonzero mass, then lower-energy neutrinos should move comparatively slowly. The data are consistent with a $E_\nu = 10\,\mathrm{MeV}$ requiring at most about $\Delta t = 10\,\mathrm{s}$ more than a photon would require to travel from the supernova to us. Find the upper limit that this observation sets on the mass of a neutrino. (Other evidence sets an even tighter limit).

  Solution
  The source is 170 kly away, meaning that photons take 170 ky to reach earth at the speed of light. The neutrinos, up to experimental uncertainty were at most 10 s slower, meaning their velocity was

  $$v \geq c\frac{170\,\mathrm{ky}}{170\,\mathrm{ky} + 10\,\mathrm{s}} = c\frac{5.364 \times 10^{12}\,\mathrm{s}}{5.364 \times 10^{12}\,\mathrm{s} + 10\,\mathrm{s}} \approx c\left(1 - \frac{10}{5.364 \times 10^{12}}\right) \approx c\left(1 - 1.864 \times 10^{-12}\right),$$

  where we used the approximation for small x:

  $$\frac{1}{1+x} \approx 1 - x.$$

  Let us assume that the $E_\nu = 10\,\mathrm{MeV}$ total energy neutrinos travel at this speed, and express their mass from the relation

  $$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1-\beta^2}},$$

  where $\beta = v/c = 1 - 1.864 \times 10^{-12}$. This results in an upper estimate on the neutrino mass

  $$m = \frac{E}{c^2}\sqrt{1-\beta^2} = \frac{E}{c^2}\sqrt{2 \times 1.864 \times 10^{-12} - (1.864 \times 10^{-12})^2} \approx \frac{19.31\,\mathrm{eV}}{c^2}.$$

- **Neutrinos from supernovae / Ex. 15.36 (Serway)**
  Supernova 1987A, located about 170 000 ly from the Earth, is estimated to have emitted a burst of neutrinos carrying energy $E \sim 10^{46}\,\mathrm{J}$. Suppose the average neutrino energy was $E_\nu = 6\,\mathrm{MeV}$ and your body presented cross-sectional area $\sigma = 5000\,\mathrm{cm}^2$. To an order of magnitude, how many of these neutrinos passed through you?

  Solution
  The total energy emitted was

  $$E_{\mathrm{tot}} = 1 \times 10^{46}\,\mathrm{J} = 1 \times 10^{46}\,\mathrm{J}\frac{e}{1.602 \times 10^{-19}\,\mathrm{C}} = 6.242 \times 10^{64}\,\mathrm{eV},$$

  where we performed the unit conversion using that the e in electronvolt stands for the unit charge and $\mathrm{V} = \frac{\mathrm{J}}{\mathrm{C}}$. Hence the number of neutrinos emitted was

  $$N = \frac{E_{\mathrm{tot}}}{E_\nu} = 1.04 \times 10^{58}.$$

  Assuming that these were radiated uniformly in every direction, by the time the neutrino burst reached Earth, they were spread out on the surface of a sphere with radius $R = 170\,000\,\mathrm{ly}$. Hence the neutrino flux reaching us was

  $$\Phi = \frac{N}{4\pi R^2} = \frac{1.04 \times 10^{58}}{4\pi(9.461 \times 10^{15}\,\mathrm{m})^2} = 9.246 \times 10^{18}\,\mathrm{m}^{-2}.$$

  The number of neutrinos passing through your body is the flux multiplied by the cross section

  $$N_{\mathrm{you}} = \Phi\sigma = 9.246 \times 10^{18}\,\mathrm{m}^{-2} \times 0.5\,\mathrm{m}^2 = 4.623 \times 10^{18}.$$

- **Ultrarelativistic de Broglie wavelength: Probing a neutron / Ex. 14.4 (Thorton)**
  To probe another particle with linear dimensions D, the wavelength of the probing particle must be $\lambda \leq D$. To learn details, the wavelength should be substantially less, perhaps as small as $0.10D$. Calculate the kinetic energies of an electron and a proton needed to probe the details of a neutron. Assume the diameter of the neutron is $r_n = 1.5\,\mathrm{fm}$.

  Solution
  The condition to probe the neutron is

  $$\lambda = \frac{h}{p} \leq 0.1\,r_n = 0.15\,\mathrm{fm} \qquad \Rightarrow \qquad p \geq \frac{h}{0.1\,r_n} = \frac{hc}{0.1\,r_n}\frac{1}{c} \approx \frac{1240\,\mathrm{eV\,nm}}{0.15\,\mathrm{fm}}\frac{1}{c} \approx 8.267\,\mathrm{GeV}/c.$$

  The kinetic energy can be derived from the total energy as

  $$E^2 = \left(K + mc^2\right)^2 = (pc)^2 + \left(mc^2\right)^2$$

  so that

  $$K = \sqrt{(pc)^2 + (mc^2)^2} - mc^2.$$

  Therefore, for an electron, $m_{\mathrm{e}} = 0.511\,\mathrm{MeV}$

  $$K_{\mathrm{e}} = 8.26\,\mathrm{GeV}$$

  and for the proton $m_{\mathrm{p}} = 938.27\,\mathrm{MeV}$

  $$K_{\mathrm{p}} = 7.38\,\mathrm{GeV}$$

  where $hc = 1239.8 \times 10^{-9}\,\mathrm{eV\,m}$ was used. Note that despite the significant mass difference $m_{\mathrm{p}}/m_{\mathrm{e}} \sim 2000$, the kinetic energy required to investigate the structure of the neutron is quite similar for the two probes

- **Ultrarelativistic de Broglie wavelength: probing shorter lengthscales / Ex. 14.5 (Thorton)**
  Some details of elementary particles may need to be probed on distance scales as small as $10^{-18}\,\mathrm{m}$. Calculate the kinetic energies of an electron and a proton necessary to probe details this small. *Hint: See previous problem, but we don't need to divide by the factor of 10 to determine the wavelength.*

  Solution
  In this case, the condition on the wavelength

  $$\lambda = \frac{h}{p} \leq 10^{-18}\,\mathrm{m}.$$

  Then the corresponding kinetic energy is given by

  $$K_\mathrm{e} = 1.24\,\mathrm{TeV}$$

  and

  $$K_\mathrm{p} = 1.24\,\mathrm{TeV}$$

  where $hc = 1239.8 \times 10^{-9}\,\mathrm{eV\,m}$ was used. Note that in the limit $K \gg mc^2$, the mass of the probe becomes irrelevant.

- **Lepton, baryon number conservation / Ex. 14.8 (Thornton)**
  Supply the missing neutrinos in the following reactions or decays

  (a) $\mu^+ \to \mathrm{e}^+ + ?$
  (b) $? + \mathrm{p} \to \mathrm{n} + \mathrm{e}^+$
  (c) $\pi^- \to \mu^- + ?$
  (d) $\mathrm{K}^- \to \mu^- + ?$
  (e) $? + \mathrm{n} \to \mathrm{p} + \mu^-$

  Solution
  The missing neutrino and the corresponding explanation are given by

  (a) $\mu^+ \to \mathrm{e}^+ + \nu_\mathrm{e} + \bar{\nu}_\mu$
  The lepton number in the electronic ($L_\mathrm{e} = 0$) and muonic ($L_\mu = -1$) flavors must be conserved. Therefore, on the right-hand-side, the presence of a positron is balanced by an electronic neutrino, and the disappearance of the antimuon is balanced by the muonic antineutrino.
  Note that the electric charge ($C = 1$) is also conserved in the process.

  (b) $\bar{\nu}_\mathrm{e} + \mathrm{p} \to \mathrm{n} + \mathrm{e}^+$
  The lepton number in the electronic flavor ($L_\mathrm{e} = -1$) must be conserved. Therefore, on the left-hand-side, the electronic antineutrino balances the position ion the right-hand-side.
  Note that the baryon number ($B = 1$) and the electric charge ($C = 1$) are also conserved in the process.

  (c) $\pi^- \to \mu^- + \bar{\nu}_\mu$
  The lepton number in the muonic flavor ($L_\mathrm{e} = 0$) must be conserved. Therefore, on the right-hand-side, the muonic antineutrino balances the muon.
  Note that the electric charge ($C = -1$) is also conserved in the process.

  (d) $\mathrm{K}^- \to \mu^- + \bar{\nu}_\mu$
  The lepton number in the muonic flavor ($L_\mathrm{e} = 0$) must be conserved. Therefore, on the right-hand-side, the muonic antineutrino balances the muon.
  Note that the electric charge ($C = -1$) is also conserved in the process.

  (e) $\nu_\mu + \mathrm{n} \to \mathrm{p} + \mu^-$
  The lepton number in the muonic flavor ($L_\mathrm{e} = 1$) must be conserved. Therefore, on the left-hand-side, the muonic neutrino balances the muon on the right-hand-side.
  Note that the baryon number ($B = 1$) and the electric charge ($C = 0$) are also conserved in the process.

- **Matter-antimatter thruster / Ex. 15.37 (Serway)**
  A rocket engine for space travel using photon drive and matter–antimatter annihilation has been suggested. Suppose the fuel for a short-duration burn consists of N protons and N antiprotons, each with mass m_p.

  (a) Assume all of the fuel is annihilated to produce photons. When the photons are ejected from the rocket, what momentum can be imparted to it?

  (b) If half of the protons and antiprotons annihilate each other and the energy released is used to eject the remaining particles, what momentum could be given to the rocket? Which scheme results in the greatest change in speed for the rocket?

  (c) Transferring from low Earth orbit to low Mars orbit requires approximately $\Delta v = 6000\,\mathrm{m/s}$. How much matter–antimatter fuel (by mass) would a 1000 t spaceship need to make a roundtrip, assuming 100% efficiency?

  Solution

  (a) Most of the energy produced by the annihilation process is the one "stored" in the proton and antiproton rest masses. Therefore, the energy per particle is given by

  $$E_\mathrm{p} = \sqrt{(pc)^2 + (m_\mathrm{p}c^2)^2} \geq m_\mathrm{p}c^2.$$

  For simplicity we assume that the particles are at rest before the annihilation, so the energy released by a proton-antiproton pair is $E = 2m_\mathrm{p}c^2$. The same energy is released in the form of two photons momentum q

  $$E_\gamma = 2|p|c,$$

  which is obtained considering that conservation of momentum requires the photons to have momenta p and $-p$ (see also the corresponding Feynman diagram above).

  Therefore, for N pairs, the total energy released is given by

  $$E = 2NE_\gamma = 2N|p|c = 2N \cdot E_\mathrm{p} = 2Nm_\mathrm{p}c^2,$$

  so that the momentum of the generated photons (assuming the photon jet to be collimated in the same direction for maximum propulsion) is given by

  $$N|p| = Nm_\mathrm{p}c.$$

  (b) If the momentum generated by the annihilation of a $(p, \bar{p})$ pair is used to propel another pair, then the kinetic energy imparted on each particle is exactly the rest energy of another annihilated particle:

  $$K = m_\mathrm{p}c^2.$$

  substituting the formula for the kinetic energy:

  $$K = E - m_\mathrm{p}c^2 = \sqrt{(pc)^2 + (m_\mathrm{p}c^2)^2} - m_\mathrm{p}c^2 = m_\mathrm{p}c^2,$$

  which results in the momentum for one ejected particle as

  $$p = \sqrt{3}m_\mathrm{p}c.$$

  Using again 2N particles in total, annihilating N and ejecting N, we get a total momentum transfer

  $$Np = \sqrt{3}Nm_\mathrm{p}c.$$

  This means that the second method of propulsion is more efficient, which is a result of massive particles having lower kinetic energy for a given momentum than massless particles.

  (c) The total momentum change needed for the trip is $\Delta p_\mathrm{tot} = 2\,\Delta v\, m_\mathrm{ship} = 1.2 \times 10^{10}\,\mathrm{kg\,m/s}$. Using the more efficient propulsion method, the total momentum transfer is related to the fuel mass as $\Delta p_\mathrm{tot} = \sqrt{3}m_\mathrm{fuel}c$. So the amount of fuel needed for the trip is

  $$m_\mathrm{fuel} = \frac{\Delta p_\mathrm{tot}}{\sqrt{3}c} = 23.09\,\mathrm{kg}.$$

- **Accelerators / Ex 14.30 (Thornton)**
  CERN constructed the first proton-proton collider, called the Intersecting Storage Rings (ISR), in 1971. Each proton beam had a kinetic energy of $K = 31\,\text{GeV}$. Calculate how much energy a fixed-target accelerator would need to have the same energy available in the center-of-mass energy.

  Solution
  To solve this problem we use the fact that the invariant length of the four-momentum is the same in every reference frame. The four-momentum is given by

  $$P = (E, pc)$$

  where E is the total energy and p is the linear momentum. The invariant length is calculated using the Minkowsky metric:

  $$|P|^2 = E^2 - (pc)^2.$$

  We apply this to the total energy and momentum of the system of the two particles in the lab frame of a fixed target experiment, assuming that particle 1 is incoming with total energy $E_1 = K_1 + mc^2$ and momentum p_1 and particle 2 is at rest. The total energy is given by

  $$E = E_1 + E_2 = E_1 + mc^2,$$

  and the total momentum by

  $$p = p_1 + p_2 = p_1.$$

  Substituting into the formula for the invariant length of the four-momentum

  $$|P|^2 = \left(E_1 + mc^2\right)^2 - (p_1 c)^2 = E_1^2 - (p_1 c)^2 + 2E_1 mc^2 + m^2c^4 = 2E_1 mc^2 + 2m^2c^4,$$

  where we used the energy-momentum relation $E_1^2 - (p_1 c)^2 = m^2c^4$.

  Now we write down the invariant length of the total four-momentum in the center-of-mass frame, where the total momentum $p_{\text{CoM}} = 0$:

  $$|P|^2 = E_{\text{CoM}}^2 - (p_{\text{CoM}} c)^2 = E_{\text{CoM}}^2.$$

  So the total energy in the center-of-mass frame is given by

  $$E_{\text{CoM}} = \sqrt{2E_1 mc^2 + 2m^2c^4} = \sqrt{2K_1 mc^2 + 4m^2c^4}.$$

  This proves Eq. (14.10) in Thorton for the case $m_1 = m_2 = m$.

  In the problem the total energy in the center-of-mass system is given for the head-on collision, $E_{\text{CoM}} = 2K + 2m_{\text{p}}c^2 = 63.88\,\text{GeV}$. Expressing the kinetic energy for a fixed target collider

  $$K_1 = \frac{E_{\text{CoM}}^2}{2m_{\text{p}}c^2} - 2m_{\text{p}}c^2 = 2.1762\,\text{TeV},$$

  two orders of magnitude higher than the ISR.